

Notes: Collado, Ortuño-Ortín & Stuhler (RES, 2023)
Estimating Intergenerational and Assortative Processes in Extended Family Data

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1 Big picture

- **Question.** How much socioeconomic advantage is transmitted across generations, and how much of that persistence comes from intergenerational transmission, sibling/common-family components, and assortative mating?
- **Core idea.** Instead of only comparing parents, grandparents, and children vertically, the paper compares many relatives within the same generation: siblings, cousins, in-laws, and very distant siblings-in-law.
- **Why horizontal kinship matters.** Horizontal relatives can be measured in the same administrative data, at similar ages and dates, and do not require observing very distant ancestors.
- **Data.** The empirical application uses Swedish multigenerational registers: a random 35% draw of the population born 1932–1967, linked to biological parents, siblings, spouses/partners, and children.
- **Main outcome.** The baseline outcome is years of schooling; the paper also studies income and height.
- **Main finding.** Latent advantages are much more persistent than conventional parent–child or sibling correlations suggest.
- **Assortative mating result.** Observable spousal correlation in education is about 0.49, but the implied spousal correlation in latent advantages is about 0.75.
- **Genetic/non-genetic result.** A standard genetic model cannot fit the wide set of kinship correlations. A two-factor model implies only about 7% of variation in schooling is explained by latent genetic factors, while non-genetic cultural factors explain much more.
- **Economic takeaway.** The observed education of parents is an imperfect proxy for family background. Hidden family advantages and hidden assortative matching generate much slower decay of kinship similarity than standard mobility measures imply.

2 Incremental contribution of this paper

- **New empirical object:** the paper turns horizontal kinship correlations into a tool for estimating intergenerational mobility. Prior work mainly used vertical links across generations or surname averages; this paper shows that same-generation relatives can reveal the missing vertical structure.
- **New data design:** by chain-linking biological siblings and spouses, the authors construct 141 distinct kinship moments, including distant affine relationships such as siblings-in-law of siblings-in-law.
- **More identifying moments:** the large set of kinship moments gives enough restrictions to estimate a model with direct observable transmission, latent transmission, sibling components, assortative mating in observables, assortative mating in latent advantages, and gender-specific effects.
- **New interpretation of assortative mating:** the paper shows that sorting on observed education is too weak to explain the similarity of distant in-laws. The relevant sorting is largely sorting on latent advantages that affect child outcomes.
- **Bridge between mobility and genetics:** unlike standard behavioral-genetics models, the paper does not force the latent factor to be genetic. It first measures total transferable advantage and then asks whether a genetic model can explain it.
- **Power of affine kin:** in-laws are crucial because they do not share ancestors but are connected through repeated assortative matches. Their correlations isolate the assortative component more sharply than vertical ancestor correlations.
- **Policy contribution:** the paper implies that measures based only on parent education, income, or sibling correlations can understate inequality of opportunity because many relevant family-background variables are latent.

3 Data and measurement

3.1 Swedish multigenerational registers

- The core sample comes from a random 35% draw of the Swedish population born between 1932 and 1967.
- Individuals are linked to biological parents, siblings, spouses/partners, and children.
- A man and woman are treated as spouses if they have a child together, whether married or cohabiting.
- The baseline child cohorts for education are born 1966–1976; parent cohorts cover 1920–1962.
- Education is measured as years of schooling from registers; income is log ten-year average pre-tax income; height comes from military enlistment data and is observed for men.

Interpretation. The data are unusually well suited for horizontal kinship because the same registers measure many relatives at similar ages and dates. This reduces the comparability problem that plagues long vertical genealogies.

3.2 Kinship moments

- The paper constructs correlations in standardized outcomes for many types of biological and affine kin.
- Consanguine examples include spouses, siblings, cousins, parent–child, uncle/aunt–child, and nephew/niece relationships.
- Affine examples include siblings-in-law, spouses of siblings-in-law, siblings-in-law of siblings-in-law, and longer chains up to fifth-order affinity.
- Gender combinations are kept separate, which expands the number of empirical moments and lets the model allow gender-specific transmission parameters.
- The final data set includes 141 kinship moments, of which 105 are used in the baseline education calibration.

Interpretation. A kinship moment is not only a descriptive statistic. Each moment places a different restriction on the same underlying transmission process. Close kins identify direct and sibling components; distant in-laws discipline latent assortative matching.

4 Model and calibration

4.1 Outcome equation

$$y_t^k = \beta^k \tilde{y}_{t-1}^k + z_t^k + x_t^k + u_t^k. \quad (1)$$

Interpretation. The coefficient β captures direct transmission through observed parental outcomes. The latent factor z captures family advantages that are not directly visible in the measured outcome but that affect descendants.

4.2 Latent transmission

$$z_t^k = \gamma^k \tilde{z}_{t-1}^k + e_t^k + v_t^k, \quad (2)$$

$$\tilde{z}_{t-1}^k = \alpha_z^k z_{t-1}^m + (1 - \alpha_z^k) z_{t-1}^f. \quad (3)$$

Interpretation. The latent status of the child is a weighted average of parental latent status plus a latent sibling component and an idiosyncratic component. The parameter γ is the key persistence parameter for hidden family advantage.

4.3 Assortative mating

$$z_{t-1}^f = r_{zz}^m z_{t-1}^m + r_{zy}^m y_{t-1}^m + w_{t-1}^m, \quad (4)$$

$$y_{t-1}^f = r_{yz}^m z_{t-1}^m + r_{yy}^m y_{t-1}^m + \varepsilon_{t-1}^m. \quad (5)$$

Interpretation. These projections separate sorting in latent status from sorting in observed schooling. The core empirical claim is that observed sorting is not enough; sorting in latent advantages is much stronger.

4.4 Calibration objective

$$\min_v \sum_{i \in C} p_i (\rho_i(v) - \hat{\rho}_i)^2. \quad (6)$$

Interpretation. The model is disciplined by the entire pattern of correlations across close, distant, consanguine, and affine kins.

5 Identification and empirical design

- Close biological kin identify direct intergenerational and sibling components because they share parents and family environments.
- Parent–child and grandparent-style correlations discipline how fast advantages travel vertically across generations.
- Cousins and uncle/aunt links provide intermediate horizontal restrictions: they share ancestors but less immediate household experience.
- Siblings-in-law are especially valuable because they do not share blood ancestry but are connected through marriage. Their similarity reveals assortative matching.
- Longer chains of in-laws make the decay of assortative matching visible. If correlations decay too slowly, observable education sorting cannot be the full explanation.
- Placebo kin from the same municipality test whether geography alone can explain the correlations. It cannot: placebo correlations are small relative to actual kin correlations.
- Restricted models show which mechanisms are necessary. Removing latent transmission, removing shared sibling components, or allowing assortative mating only in observables materially worsens fit.
- Genetic model comparison uses the same kinship moments to ask whether a standard genetic model can reproduce the correlation pattern. It fits close relatives better than distant in-laws, so it fails the full horizontal test.

Interpretation. The model is a reduced-form decomposition, not a randomized causal experiment. The paper identifies which transmission processes are required to fit kinship correlations, not the exact causal mechanism behind each latent component.

6 Tables and figures

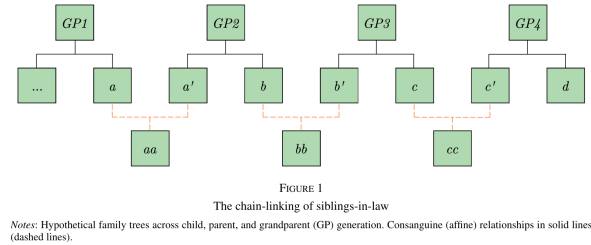
6.1 Figure 1 + Table 2: Horizontal kinship and the kinship menu

Read: Figure 1 shows how repeated sibling and spouse links generate long chains of in-laws. Solid lines represent consanguine links; dashed lines represent affine links through spouses. The key move is that distant in-laws can be identified without observing distant ancestors.

Read: Table 2 lists examples of kinship groups and the number of gender-specific moments in each group. Spouses generate one moment; siblings generate three; cousins generate ten; parent–child links generate four. Higher-order in-law chains create many more moments once gender combinations are separated.

Interpretation. The figure and table together explain the paper’s methodological innovation. Affine chains turn marriage links into a measurement device for sorting; biological links turn family trees into restrictions on vertical transmission. The combination identifies mechanisms that would be confounded in vertical data alone.

Economic message. Horizontal kinship is not a curiosity. It is the source of identification for latent assortative matching and for the persistence of family advantage beyond observed education.



(a) Figure 1

TABLE 2
List of kinship types

Group	Name	Class	No. of moments
$a-a'$	Spouses	Consanguine, horizontal	1
$a'-b$	Siblings	Consanguine, horizontal	3
$aa-bb$	Cousins	Consanguine, horizontal	10
$aa-a$	Parent-child	Consanguine, vertical	4
$aa-b$	Uncle/aunt-child	Consanguine, horizontal, and vertical	8
$a-b$	Siblings-in-law (degree 1)	Affine, horizontal	4
$a-b'$	Spouse of sib-in-law (degree 1)	Affine, horizontal	3
$a'-c$	Sibling of sib-in-law (degree 1)	Affine, horizontal	4
$aa-b'$	Child-sibling in law (degree 1)	Affine, horizontal, and vertical	8
$a-c$	Siblings-in-law (degree 2)	Affine, horizontal	8
...	...	...	...
$a-d$	Siblings-in-law (degree 3)	Affine, horizontal	16
...	...	...	...

(b) Table 2

6.2 Table 1: Descriptive statistics

Read: The education sample contains more than one million child observations and more than 2.5 million parent observations. Women have slightly higher average years of schooling than men among child cohorts. Parent cohorts have lower education, reflecting education expansion across generations.

Interpretation. The table shows why Sweden is useful for this design: the authors have massive administrative samples and can observe multiple outcomes for both parents and children. The education expansion across cohorts makes standardization by cohort and gender essential.

Economic message. The data are rich enough to estimate hundreds of kinship correlations precisely.

TABLE 1
Descriptive statistics in Swedish registers

Variable	Generation	Cohorts	Observations	Men		Women	
				Mean	Std. Dev.	Mean	Std. Dev.
Education	Child	1966–76	1,026,084	12.35	2.16	12.81	2.12
	Parent	1920–62	2,502,869	10.70	3.01	10.60	2.88
Income	Child	1958–68	1,062,847	12.30	0.55	11.97	0.50
	Parent	1923–53	1,811,895	12.37	0.52	11.82	0.80
Height	Child	1974–80	269,475	179.73	6.55		
	Parent	1951–66	378,686	178.99	6.37		

Notes: Education is measured as years of education. Income is measured as the logarithm of 10-year average pre-tax incomes at age 30–39 for children and age 45–54 for parents. Height in cm is observed for men only.

6.3 Table 3: Sample and predicted kinship moments

Read: Spousal schooling correlation is about 0.491, and sibling correlations are around 0.432 for brothers, 0.416 for sisters, and 0.375 for brother–sister pairs. Parent–child correlations are lower but still substantial. Sibling-in-law correlations remain high, around 0.28–0.30 for first-degree in-laws. More distant in-law and cousin correlations decline but do not collapse to zero.

Interpretation. Table 3 is the empirical backbone of the paper. The slow decline of correlations across in-law chains is especially informative: in-laws do not share ancestors, so their similarity must come through assortative matching and the persistence of traits selected in spouses.

Economic message. Inequality transmission is visible not only in parent–child links but across a network of relatives. The persistence of similarity among distant relatives implies a stronger hidden family-background process than conventional mobility statistics suggest.

6.4 Figure 2 + Table 4: Baseline fit and calibrated parameters

Read: Figure 2 shows that the model captures the broad downward pattern from close to distant kin. Table 4 shows strong latent transmission, small direct effects of parental education, high sibling correlations in latent status, and an implied spousal correlation in latent status of about 0.754.

Interpretation. Figure 2 establishes that the model can fit many moments simultaneously. Table 4 then explains how: direct transmission through observed education is modest, while latent transmission, sibling components, and latent spousal sorting are strong.

Economic message. The family background that matters for long-run mobility is not fully visible in education registers.

6.5 Figure 3 + Figure 4: Extrapolation and restricted models

Read: Figure 3 shows that a model calibrated on a smaller set of close moments predicts many excluded distant moments. Figure 4 shows that removing latent transmission, shared sibling components, or latent assortative mating worsens fit.

Interpretation. These figures are mechanism tests. Figure 3 asks whether the estimated model can extrapolate. Figure 4 asks which components are indispensable. The answer is that latent transmission, sibling-specific components, and latent assortative mating are all needed.

TABLE 3
Sample and predicted moments in Swedish registers

Group #	Kinship		Data		Calibration	
	No.	Name	No. of pairs	Sample ρ	Pred. ρ	% error
I	1	Spouses	413,062	0.491	0.489	-0.3
	2	Brothers	387,028	0.432	0.431	-0.3
	3	Sisters	431,698	0.416	0.416	0.2
II	4	Brother-Sister	800,127	0.375	0.376	0.4
	5	Father-Son	396,304	0.380	0.381	0.2
	6	Father-Daughter	376,255	0.321	0.321	0.1
III	7	Mother-Son	422,374	0.366	0.367	0.2
	8	Mother-Daughter	400,337	0.347	0.349	0.5
	9	Brothers-in-law (MF-M)	602,262	0.302	0.296	-2.1
IV	10	Brother-Sister-in-law (FM-M)	578,269	0.296	0.304	2.7
	11	Brother-Sister-in-law (MF-F)	650,127	0.298	0.307	3.0
	12	Sisters-in-law (FM-F)	596,540	0.278	0.277	-0.2
V	13	Nephew-Uncle (BF)	280,967	0.254	0.249	-1.7
	14	Niece-Uncle (BF)	266,289	0.218	0.220	1.2
	15	Nephew-Uncle (BM)	312,019	0.241	0.238	-1.2
VI	16	Niece-Uncle (BM)	295,580	0.209	0.210	0.5
	17	Nephew-Aunt (SF)	285,618	0.234	0.229	-2.1
	18	Niece-Aunt (SF)	270,325	0.217	0.203	-5.5
VII	19	Nephew-Aunt (SM)	333,141	0.251	0.245	-2.3
	20	Niece-Aunt (SM)	316,625	0.234	0.218	-7.0
	21	Spouse of Sib-in-law (MF-FM)	252,232	0.252	0.246	-2.2
VIII	22	Spouse of Sib-in-law (FM-MF)	226,795	0.229	0.232	1.1
	23	Spouse of Sib-in-law (MF-MF)	464,081	0.227	0.227	1.9
	24	Nephew-Aunt-in-law (BF)	231,767	0.192	0.192	-0.3
IX	25	Niece-Aunt-in-law (BF)	221,287	0.172	0.171	-0.8
	26	Nephew-Aunt-in-law (BM)	254,534	0.187	0.183	-2.2
	27	Niece-Aunt-in-law (BM)	241,873	0.164	0.161	-2.0
X	28	Nephew-Uncle-in-law (SF)	227,403	0.190	0.188	-1.5
	29	Niece-Uncle-in-law (SF)	215,968	0.163	0.167	2.2
	30	Nephew-Uncle-in-law (SM)	264,524	0.197	0.198	0.8
XI	31	Niece-Uncle-in-law (SM)	251,782	0.171	0.175	2.1
	32	Male Cousins (BF)	70,137	0.208	0.159	-23.8
	33	Male Cousins (SM)	82,049	0.215	0.160	-25.4
XII	34	Male Cousins (SF)	156,747	0.202	0.152	-24.8
	35	Female Cousins (BF)	63,012	0.169	0.126	-25.5
	36	Female Cousins (SM)	71,649	0.197	0.124	-37.0
XIII	37	Female Cousins (SF)	140,522	0.177	0.118	-33.2
	38	Male-Female Cousins (BF)	144,100	0.179	0.141	-20.9
	39	Male-Female Cousins (SM)	170,577	0.196	0.141	-28.2
XIV	40	Male-Female Cousins (BM)	148,691	0.179	0.133	-25.5
	41	Male-Female Cousins (SM)	148,631	0.184	0.135	-26.5
	42	F-M-F	461,863	0.185	0.191	3.5
XV	43	F-M-F	500,448	0.192	0.196	1.8
	44	F-M-F	481,006	0.207	0.212	2.6
	45	M-F-M	447,263	0.208	0.207	-0.6
XVI	46	M-F-M	962,409	0.156	0.157	0.8
	47	M-F-M	393,579	0.156	0.161	3.4
	48	M-F-M	375,442	0.179	0.173	-3.4
XVII	49	M-F-M	391,389	0.158	0.160	1.2
	50	M-F-M	353,470	0.160	0.161	0.8
	51	M-F-M	378,720	0.164	0.165	0.5
XVIII	52	M-F-M	341,316	0.157	0.160	2.1
	53	M-F-M	351,350	0.148	0.148	0.0
	54	M-F-M	202,632	0.119	0.127	6.7
XIX	55	M-F-M	219,007	0.129	0.130	1.1
	56	M-F-M	192,819	0.134	0.140	4.6
	57	M-F-M	199,811	0.129	0.129	-0.3
XX	58	M-F-M	183,670	0.125	0.133	6.2
	59	M-F-M	196,631	0.132	0.136	3.6
	60	M-F-M	160,857	0.132	0.132	0.6
XXI	61	M-F-M	164,528	0.122	0.122	-0.2
	62	F-M-F	192,818	0.104	0.112	7.1
	63	F-M-F	208,008	0.114	0.114	0.2
XXII	64	F-M-F	183,929	0.116	0.123	6.1
	65	F-M-F	191,177	0.113	0.113	1.2
	66	F-M-F	175,507	0.110	0.119	8.0
XXIII	67	F-M-F	187,178	0.121	0.122	0.5
	68	F-M-F	151,006	0.112	0.118	5.5
	69	F-M-F	155,658	0.113	0.109	-3.7
XXIV	70	M-F-M	278,938	0.122	0.122	-0.5
	71	M-F-M	310,160	0.132	0.132	-0.6
	72	M-F-M	299,602	0.138	0.135	-2.2
	73	M-F-M	273,809	0.126	0.124	-1.4

(Continued)

(a) Table 3, part 1

TABLE 3
(Continued)

Group	Kinship		Data		Calibration	
	No.	Name	No. of pairs	Sample ρ	Pred. ρ	% error
XIII	74	M-M-FM-FM	160,726	0.102	0.098	-3.5
	75	M-M-FM-FM	174,261	0.103	0.106	3.4
	76	M-M-FM-FM	158,401	0.107	0.109	1.9
XIV	77	M-M-FM-FM	160,105	0.106	0.106	0.3
	78	M-M-FM-FM	147,949	0.102	0.103	0.9
	79	M-M-FM-FM	156,876	0.103	0.111	7.9
XV	80	M-M-FM-FM	133,588	0.104	0.103	-1.0
	81	M-M-FM-FM	131,756	0.101	0.101	-0.5
	82	M-M-FM-FM	153,751	0.087	0.086	-0.1
XVI	83	F-M-FM-FM	165,828	0.091	0.093	3.1
	84	F-M-FM-FM	151,100	0.094	0.095	1.7
	85	F-M-FM-FM	153,065	0.089	0.091	4.9
XVII	86	F-M-FM-FM	140,585	0.093	0.092	-1.1
	87	F-M-FM-FM	150,162	0.097	0.099	2.9
	88	F-M-FM-FM	126,129	0.093	0.092	-1.2
XVIII	89	F-M-FM-FM	124,968	0.085	0.090	5.3
	90	M-M-FM-FM → M	84,025	0.094	0.082	-13.4
	91	M-M-FM-FM → M	100,261	0.101	0.086	-15.2
XIX	92	M-M-FM-FM → M	93,237	0.105	0.090	-15.9
	93	M-M-FM-FM → M	80,486	0.097	0.085	-12.0
	94	M-M-FM-FM → F	79,090	0.087	0.073	-16.7
XX	95	M-M-FM-FM → F	95,733	0.094	0.076	-19.9
	96	M-M-FM-FM → F	89,364	0.093	0.080	-13.6
	97	M-M-FM-FM → F	95,020	0.095	0.075	-20.4
XXI	98	M-M-FM-FM → F	76,514	0.095	0.076	-20.0
	99	M-M-FM-FM → F	89,054	0.088	0.079	-9.8
	100	M-M-FM-FM → F	77,532	0.094	0.076	-19.0
XXII	101	M-M-FM-FM → F	80,867	0.082	0.072	-12.9
	102	F-M-FM-FM → F	76,344	0.080	0.064	-20.3
	103	F-M-FM-FM → F	91,080	0.090	0.066	-26.6
XXIII	104	F-M-FM-FM → F	84,736	0.092	0.070	-23.4
	105	F-M-FM-FM → F	72,410	0.082	0.068	-17.4
	106	F-M-FM-FM	288,374	0.103	0.103	0.2
XXIV	107	F-M-FM-FM	312,703	0.102	0.105	3.5
	108	F-M-FM-FM	311,795	0.111	0.113	1.4
	109	F-M-FM-FM	162,928	0.099	0.104	5.4
XXV	110	F-M-FM-FM	108,163	0.116	0.115	-1.5
	111	F-M-FM-FM	166,250	0.121	0.117	-2.6
	112	F-M-FM-FM	304,684	0.114	0.114	0.3
XXVI	113	M-M-FM-FM	278,416	0.117	0.111	-4.9
	114	M-M-FM-FM	149,478	0.131	0.122	-7.0
	115	M-M-FM-FM	143,733	0.115	0.112	-2.2
XXVII	116	M-M-FM-FM	230,313	0.087	0.085	-2.5
	117	M-M-FM-FM	251,223	0.087	0.087	0.2
	118	M-M-FM-FM	248,811	0.097	0.093	-4.0
XXVIII	119	M-M-FM-FM	259,925	0.083	0.086	3.0
	120	M-M-FM-FM	245,814	0.104	0.094	-9.7
	121	M-M-FM-FM	265,220	0.101	0.096	-4.5
XXIX	122	M-M-FM-FM	241,998	0.099	0.093	-6.0
	123	M-M-FM-FM	248,449	0.084	0.086	1.7
	124	M-M-FM-FM	224,873	0.091	0.087	-5.1
XXX	125	M-M-FM-FM	246,186	0.092	0.089	-3.9
	126	M-M-FM-FM	237,791	0.100	0.095	-5.1
	127	M-M-FM-FM	247,495	0.088	0.088	0.0
XXXI	128	M-M-FM-FM	223,661	0.086	0.087	0.3
	129	M-M-FM-FM	240,328	0.094	0.094	0.0
	130	M-M-FM-FM	213,155	0.091	0.086	-5.5
XXXII	131	M-M-FM-FM	220,553	0.084	0.079	-4.9
	132	M-M-FM-FM	176,790	0.071	0.066	-8.3
	133	M-M-FM-FM	199,041	0.075	0.071	-6.0
XXXIII	134	M-M-FM-FM	153,057	0.047	0.046	-3.3
	135	M-M-FM-FM	144,976	0.054	0.043	-20.1
	136	M-M-FM-FM	117,473	0.047	0.045	-2.5
XXXIV	137	M-M-FM-FM	135,096	0.042	0.038	-9.9
	138	M-M-FM-FM	106,844	0.031	0.025	-21.9
	139	M-M-FM-FM	100,871	0.043	0.021	-46.5
XXXV	140	M-M-FM-FM	82,523	0.032	0.019	-40.0
	141	M-M-FM-FM	96,840	0.027	0.021	-24.4

Notes: MF, indicate gender, MF or FM are spouses, - indicates a sibling link and → indicates a parent-child link. BF, brother of the father; BM, brother of the mother; SF, sister of the father; SM, sister of the mother. Moments printed in italics were not used in the calibration. The sample correlations for cousins are subject to limitations discussed in Section 4.6.

(b) Table 3, part 2

TABLE 4
Calibrated parameters in Swedish registers (years of schooling)

Panel A: Intergenerational processes						
β^m	β^f	γ^m	γ^f	σ_{γ}^2	σ_{γ}^2	σ_{γ}^2
0.144	0.129	0.664	0.566	4.648	4.465	2.072
Panel B: Sibling processes						
σ_{γ}^2	σ_{γ}^2	σ_{γ}^2	σ_{γ}^2	σ_{γ}^2	σ_{γ}^2	σ_{γ}^2
0.180	0.244	0.068	0.657	0.712	0.625	0.711
Panel C: Assortative processes						
ρ_{γ}^m	ρ_{γ}^f	ρ_{γ}^m	ρ_{γ}^f	ρ_{γ}^m	ρ_{γ}^f	ρ_{γ}^m
0.662	0.608	0.696	0.143	0.747	0.113	0.661
Panel D: Variance decomposition of γ						
γ_{-1}	γ_{-1}	γ_{-1}	γ_{-1}	γ_{-1}	γ_{-1}	γ_{-1}
male	1.5%	16.1%	7.5%	25.1%	3.9%	43.1%
female	1.6%	12.9%	5.7%	20.2%	5.5%	41.6%

Notes: Panels A and B report parameter estimates and implied ancestor and sibling correlations in educational attainment γ and latent advantages z . The parameters β and σ_{γ}^2 capture the direct transmission and variance of γ ; γ and σ_{γ}^2 the transmission and variance of z ; and σ_{γ}^2 and σ_{γ}^2

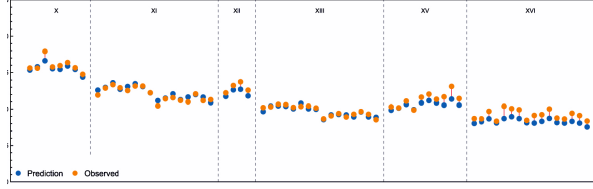


FIGURE 3
Out-of-sample fit in Swedish registers

(a) Figure 3

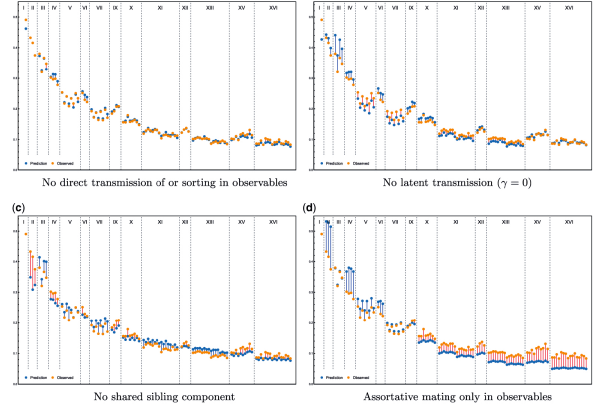


FIGURE 4
Restricted models

(b) Figure 4

6.6 Figure 5: Actual versus placebo kin

Read: Blue dots show actual kin correlations; orange dots show placebo correlations for non-kin from the same municipality and birth year. Placebo correlations are positive but much smaller than actual kin correlations.

Interpretation. If geography explained the kinship correlations, same-municipality placebo pairs would look similar to actual kins. They do not.

Economic message. Family background is not reducible to local environment. The latent factor captures broader inherited socioeconomic, cultural, and family-specific advantages.

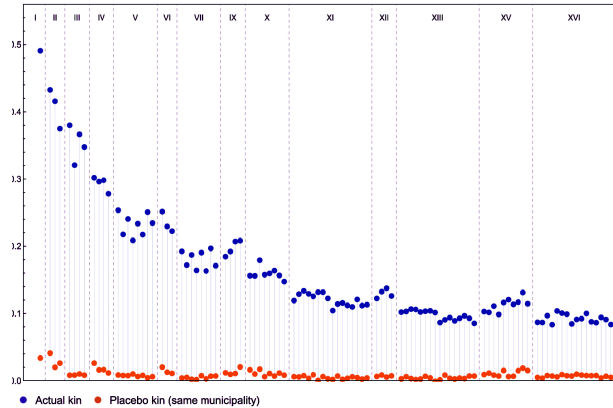


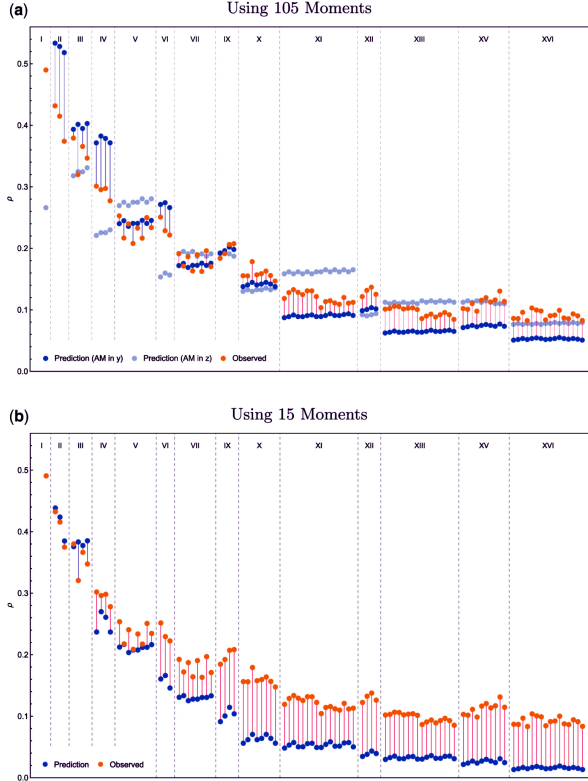
FIGURE 5
Actual vs placebo kin

6.7 Figure 6 + Table 5: Genetic model and two-factor decomposition

Read: Figure 6 shows that the standard genetic model has poor fit when confronted with 105 moments; it can look acceptable when fit only to 15 close moments but then fails for the rest. Table 5 decomposes latent status into genetic and cultural/non-genetic components.

Interpretation. Close relatives alone can make a genetic model look plausible. Distant in-laws reveal that non-genetic sorting and cultural transmission are central. The two-factor model implies that genetic factors explain about 7% of variance in schooling, while cultural/non-genetic factors explain much more.

Economic message. The hidden component of family advantage is mostly not genetic in the narrow sense. The persistence of inequality is driven largely by non-genetic family, cultural, and socioeconomic channels that are transmitted and sorted across families.



(a) Figure 6

TABLE 5
Calibrated parameters in Swedish registers (education, two factor model)

Panel A: Intergenerational Processes							
β^m	β^f	γ^m	γ^f	σ_{ym}^2	σ_y^2	σ_{zm}^2	σ_{zg}^2
0.116	0.100	0.690	0.587	4.648	4.465	1.753	1.347
σ_{ym}	σ_{yf}	σ_{zm}	σ_{zf}	σ_{ym}^2	σ_{yf}^2	σ_{zm}^2	σ_{zf}^2
0.448	0.003	0.577	0.659	2.058	2.319		
Within-person correlations in y and z:							
	ρ_{yz}	ρ_{zg}	ρ_{zcg}				
male	0.667	0.304	0.034				
female	0.594	0.304	0.034				
Ancestor correlations in y and z:							
	Father-Son	Fi-Dau.	Mother-Son	Mo-Dau.			
in zc	0.576	0.580	0.533	0.507			
in zg	0.513	0.513	0.513	0.513			
in zc+zc	0.582	0.585	0.546	0.527			
	Father-Son	Grandfr-Son	GGfr-Son	GGGfr-Son			
in y	0.379	0.209	0.121	0.071			
in zc+zc	0.582	0.340	0.199	0.116			
Panel B: Sibling Processes							
	σ_{ym}^2	σ_{yf}^2	σ_{zm}^2	σ_{zf}^2	σ_{emf}^2		
in zc	0.120	0.173	0.000	0.681	0.729	0.652	
Sibling correlations in zc and zg:							
	Brothers	Sisters	Mixed	in zg	Brothers	Sisters	Mixed
in zc	0.754	0.898	0.785		0.513	0.513	0.513
Panel C: Assortative Processes							
ρ_{ymf}	ρ_{ymf}	ρ_{ymf}	ρ_{ymf}	ρ_{ymf}	ρ_{ymf}	ρ_{ymf}	ρ_{ymf}
0.491	0.514	0.099	0.546	0.695	0.082	0.083	0.049
Panel D: Variance Decomposition of y							
	γ_{i-1}	γ_{i-1}^G	γ_{i-1}^G	Σ_{i-1}	γ_i	ϵ_i	$\Sigma_{i-1,i}$
male	1.0%	13.8%	3.5%	7.6%	25.9%	2.6%	14.6%
female	1.0%	10.8%	3.6%	6.0%	21.4%	3.9%	16.3%
							41.5%

Notes: Panels A and B report parameter estimates and implied ancestor and sibling correlations from calibration of the extended model with genetic and non-genetic latent factors (z^G and z^C). The parameters β and σ^2 capture the direct transmission and variance of y; γ and σ_z^2 the transmission and variance of z^C ; σ_{zm}^2 the variance of z^G ; and σ_y^2 and σ_z^2 shared sibling influences in y and z^C ; see equations (2)–(3) and (7)–(8) for details. Panel C reports the coefficients from the linear projections (4) and (5) of z^C and y^G for $k=f,m$ on their spouse's z and y and the implied spousal correlations. Panel D decomposes σ_y^2 into the components explained by the parents' y (γ_{i-1}), z^C (γ_{i-1}^G), z^G (γ_{i-1}^G) and their covariances.

(b) Table 5

7 Short summary

- The paper develops a horizontal-kinship approach to estimate intergenerational transmission and assortative mating.
- The empirical setting uses Swedish administrative registers with biological and spousal links.
- The method constructs 141 kinship moments, including very distant in-laws.
- The baseline model separates direct observable transmission, latent transmission, sibling components, and assortative mating.
- Observed parent education has a smaller direct role than latent family advantages.
- Sibling and spousal similarities in latent advantage are much stronger than their observed schooling correlations.

- The spousal correlation in observed education is about 0.49, but the spousal correlation in latent advantage is about 0.75.
- Geography explains only a small part of kinship correlations.
- A standard genetic model cannot explain the full pattern of kinship moments.
- A two-factor model implies roughly 7% genetic contribution to schooling variance and a much larger cultural/non-genetic contribution.

8 Expanded conclusion

8.1 Core conclusion

Collado, Ortuño-Ortín and Stuhler show that horizontal kinship moments reveal a much stronger transmission of family advantage than standard parent–child or sibling correlations suggest. By comparing many types of same-generation relatives, especially distant in-laws, the paper uncovers latent persistence and latent assortative mating that are invisible in conventional designs.

8.2 Why the contribution matters

The paper changes what can be learned from administrative registers. Rather than treating family trees only as vertical lineages, it treats the family network as a set of identifying restrictions. This lets researchers estimate richer mobility models even when distant ancestors are not observed.

8.3 Interpretation of latent advantages

The latent factor should not be interpreted as one single mechanism. It is a reduced-form index of advantages that affect descendants but are not fully measured by observed schooling, income, or height. These advantages may include parental expectations, networks, neighborhood choices, cultural practices, non-cognitive skills, wealth, school quality, social capital, and other family-level factors.

8.4 Implications for mobility research

Standard parent–child correlations can be too small not because mobility is high, but because the observed outcome is a noisy measure of the deeper advantage transmitted by families. Sibling correlations are more comprehensive, but the paper shows that even they can understate latent similarity. Future mobility work should therefore distinguish between observed mobility and mobility in latent socioeconomic prospects.

8.5 Implications for assortative mating

The paper’s strongest substantive result is the estimated strength of latent sorting. Spouses are not merely similar in education; they appear highly similar in the unobserved determinants of children’s prospects. This matters because assortative mating amplifies dynastic persistence.

8.6 Nature versus nurture

The horizontal design also offers a new way to distinguish genetic and non-genetic pathways. Vertical relatives can make these pathways look statistically similar, but affine relatives break the symmetry because

they are connected through marriage rather than blood. The paper’s evidence suggests that the bulk of latent persistence in educational attainment is non-genetic.

8.7 Policy implications

The findings imply that policies aimed only at observed parental education or income may miss important dimensions of inherited advantage. Equal-opportunity policy needs to address broader family-background channels: early-childhood environments, school quality, information, peer and neighborhood effects, social networks, and parental resources that are not captured by simple administrative outcomes.

8.8 Limitations and next steps

The model is intentionally reduced form. It does not fully separate all causal channels, does not model fertility choices, and assumes that the moments can be interpreted through a stable transmission structure. Future work can extend the approach to non-linear transmission, cohort dynamics, fertility, and cross-country comparisons.

8.9 Takeaway

Mobility is a network phenomenon, not just a parent–child statistic. Extended family data reveal hidden persistence in status, hidden sorting in marriage, and a large non-genetic component of transmitted advantage.